Linear Independence of Matrix Columns

Suppose that we begin with a matrix $A=[a_1 \ldots\ldots a_n]$ instead of a set of vectors. The matrix equation $Ax=0$ can be written as

$$\boxed{x_1a_1+x_2a_2+\ldots\ldots+x_na_n=0}$$

Each linear dependence relation among the columns of A corresponds to a nontrivial solution of $Ax=0$. Thus we have the following important fact:

> The columns of a matrix A are linearly independent if and only if the equation $Ax=0$ has only the trivial solution. (3)

Example 2: Determine if the columns of the matrix $A=\begin{bmatrix} 0 & 1 & 4 \\ 1 & 2 & -1 \\ 5 & 8 & 0 \end{bmatrix}$ are linearly independent.

Solution: By Theorem 3, the matrix equation $Ax=0$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ 0]$. By Theorem 2, since the matrix eqn. $Ax=0$ is consistent (trivial solution), the eqn. has a unique solution if and only if there are no free variables. By defn., a free variable corresponds to a column having no pivots. By defn, a pivot corresponds to a leading 1 of a row in rref.

$$\begin{bmatrix} 0 & 1 & 4 & 0 \\ 1 & 2 & -1 & 0 \\ 5 & 8 & 0 & 0 \end{bmatrix} \xrightarrow[\text{Row 1 and Row 2}]{\text{Interchange}} \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & 4 & 0 \\ 5 & 8 & 0 & 0 \end{bmatrix} \xrightarrow[-5\times \text{Row1}]{\text{Row3}=\text{Row3}} \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & -2 & 5 & 0 \end{bmatrix}$$

$$\xrightarrow[+2\times \text{Row2}]{\text{Row3}=\text{Row3}} \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 13 & 0 \end{bmatrix} \xrightarrow[1/13]{\text{Row3}=\text{Row3}\times} \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \xrightarrow[+\text{Row3}]{\text{Row1}=\text{Row1}}$$

$$\begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \xrightarrow[-4\times \text{Row1}]{\text{Row2}=\text{Row2}} \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \xrightarrow[-2\times \text{Row2}]{\text{Row1}=\text{Row1}} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$